

Key Characteristics of the Dataset

March 19, 2025

sicor_municipal_co... .csv
16.7 MB

Analyze the provided dataset to identify its key characteristics for data-driven decision-making. Upon receiving a specific analytical question, address it strictly using the dataset. Your deliverables must include a clear, concise answer to the question and the Python code that extracts and processes the data used to derive your answer.

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Key Characteristics of the Dataset

The dataset contains information about municipal contracts, with the following key fields:

1. **StateCode** : Identifier for the state.
2. **MunicipalityCode** : Identifier for the municipality.
3. **MonthIssued** : Month when the contract was issued.
4. **ProgramCode** and **SubProgramCode** : Codes identifying the program and subprogram associated with the contract.
5. **ResourceSourceCode** : Code indicating the source of resources.
6. **EconomicActivity** : Identifier for the economic activity linked to the contract.
7. **FundingQuantity** , **FundingValue** : Metrics related to funding (quantity and monetary value).
8. **InvestmentQuantity** , **InvestmentValue** : Metrics related to investment (quantity and monetary value).
9. **CommercializationQuantity** , **CommercializationValue** : Metrics related to commercialization (quantity and monetary value).
10. **IndustrializationQuantity** , **IndustrializationValue** : Metrics related to industrialization (quantity and monetary value).
11. **AllocatedAreaInvestment** , **AllocatedAreaFunding** : Allocated areas for investment and funding.
12. **FinancialInstitutionCNPJ** : Tax ID of the financial institution involved.
13. **ContractTotalValue** : Total value of the contract.

Analytical Question

To proceed, I need a specific analytical question to address. For example:

- What is the total value of contracts issued in a specific state?
- Which economic activity has the highest total funding value?
- What is the average contract value per municipality?

Please provide your analytical question, and I will answer it using the dataset along with the corresponding Python code.

Which municipality recorded the highest total investment value in each specific program?

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To determine which municipality recorded the highest total investment value for each specific program, we need to group the data by `ProgramCode` and `MunicipalityCode`, summing up the `InvestmentValue` for each group. Then, for each `ProgramCode`, we identify the municipality with the maximum total `InvestmentValue`.

Here is the Python code to extract and process the data:

```
python
1 import pandas as pd
2
```



```
3 # Load the dataset
4 file_path = 'sior_municipal_contracts.csv'
5 data = pd.read_csv(file_path)
6
7 # Group by ProgramCode and MunicipalityCode, summing the InvestmentValue
8 grouped_data = data.groupby(['ProgramCode', 'MunicipalityCode'])['InvestmentValue'].sum().reset_
9
10 # Find the municipality with the highest total investment value for each program
11 highest_investment = grouped_data.loc[grouped_data.groupby('ProgramCode')['InvestmentValue'].idx
12
13 # Display the results
14 print(highest_investment[['ProgramCode', 'MunicipalityCode', 'InvestmentValue']])
```

Explanation of the Code:

- 1. **Load the Dataset** : The dataset is loaded using `pd.read_csv` .
- 2. **Group and Aggregate** : We group the data by `ProgramCode` and `MunicipalityCode` and calculate the total `InvestmentValue` for each combination.
- 3. **Find Maximum per Program** : For each `ProgramCode` , we find the row with the maximum `InvestmentValue` using `idxmax()` .
- 4. **Output Results** : The result includes the `ProgramCode` , `MunicipalityCode` , and the corresponding maximum `InvestmentValue` .

Example Output:

The output will look something like this (values are illustrative):

PROGRAMCODE	MUNICIPALITYCODE	INVESTMENTVALUE
999	20011	253051.3
1	16977	3260000.0
50	13248	1087467.0

This table shows the municipality with the highest total investment value for each program.

Let me know if you'd like further clarification or additional analysis!

